

OISD-STD-181

Geophysical Operations Safety Standards

Industry: Oil & Natural Gas

Category: Exploration (Upstream) - Geophysics

Edition/Revision: 1st Edition / Currently Active

Research Date: August 26, 2026

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1. STANDARD OVERVIEW

Standard Number: OISD-STD-181

Official Title: Geophysical Operations

Type: Standard (STD)

Industry Category: Exploration & Production (Upstream)

Main Subject: Establishing comprehensive safety standards, procedures, and protocols for conducting onshore geophysical (seismic) operations in India. It covers the entire lifecycle of a seismic campaign, from setting up base camps to drilling shot holes, handling explosives, and recording seismic data.

2. PURPOSE OF THE STANDARD

Geophysical surveys (primarily seismic surveys) are scientific methods used for the discovery of hydrocarbons. These operations are often conducted in remote, rugged, and hostile environments.

The purpose of **OISD-STD-181** is to standardize safety practices across the Indian oil and gas industry to prevent accidents during these complex field operations. It aims to protect the survey crews from environmental hazards (like floods, wildlife), mechanical hazards (drilling rigs, heavy vibroseis trucks), and chemical/explosive hazards (handling detonators and seismic charges).

3. SCOPE

The standard covers safety aspects during the following activities:

- **Camp Setup:** Establishing safe remote camps for personnel, material storage, and vehicle parking.
- **Survey Activities:** Communication and safe physical surveying of land.
- **Drilling Operations:** Safe drilling of shot holes using mechanical rigs or manual (dheki) systems.
- **Explosives:** Storage, transportation, and usage of explosives and detonators.
- **Energy Sources:** Safety protocols for both explosive "shooting" and mechanical "Vibroseis" operations.
- **Data Acquisition:** Seismic recording operations.

Exclusions: The standard explicitly states that it acts as a supplement to, and does not supersede, statutory requirements from bodies such as the Director General of Mines Safety (DGMS) or the Chief Controller of

Explosives (CCE). It also notes that no standard can substitute for the judgment of a responsible and experienced geoscientist.

4. INDUSTRY APPLICATION

OISD-STD-181 is applied at the very beginning of the upstream lifecycle—the **Exploration Phase**. It is utilized by:

- Geoscientists and Seismologists planning land surveys.
- Seismic data acquisition contractors deploying crews to remote blocks.
- Shot hole drilling teams and explosive handlers (shooters).
- HSE officers auditing field camps and logistics.

5. MAJOR AREAS COVERED

Based on the standard's index, the major areas covered include:

- Camp Safety (Section 3)
- Communication (Section 4)
- Survey Operations (Section 5)
- Shot Hole Drilling (Section 6)
- Explosives & Shooting (Sections 7 & 8)
- Seismic Recording (Section 9)
- Vehicle Safety in the Field (Section 10)
- Vibroseis Operations (Section 11)
- Training (Section 12)

6. DETAILED TOPIC-WISE SUMMARY

Camp Safety

What it covers: Guidelines for setting up temporary bases for crew living and equipment storage.

Important Requirements: Camps must not be located in flood-prone areas. The perimeter must be properly fenced and guarded to protect against wildlife/reptiles. Safe, hygienic environments must be maintained with regular pest control (e.g., DDT, Phenyl sprays). It mandates the provision of potable

drinking water in steel or high-quality plastic containers, basic first aid training for personnel, and designated doctors for regular camp visits.

Shot Hole Drilling

What it covers: Creating small-diameter vertical holes where explosive charges will be placed.

Important Requirements: Details mechanical safeguards for operating portable drilling rigs and manual "dheki" systems, ensuring the crew is protected from moving parts and structural collapses while drilling to predetermined depths.

Explosives and Shooting

What it covers: The handling of seismic explosives used as a subsurface energy source.

Important Requirements: Dictates the strict separation of detonators from main explosive charges during storage and transport. Only authorized personnel are permitted to load explosives into shot holes and conduct the "shooting" (detonation) phase, in strict adherence to CCE guidelines.

Vibroseis Operations

What it covers: The use of large, truck-mounted vibrators that press against the earth to generate seismic waves mechanically, rather than using dynamite.

Important Requirements: Operators must manage the risks of heavy, moving machinery and high-pressure hydraulic systems. Safe exclusion zones must be maintained around the vibrator pads while they are actively shaking the ground.

7. EQUIPMENT AND SYSTEMS

Equipment / System	Purpose	Standard Requirement / Context
Shot Hole Drilling Rig	To drill vertical holes into the earth for explosive placement.	Must be operated safely to prevent mechanical entanglement or tipping in rugged terrain.
Vibroseis Trucks	Generates artificial seismic vibrations.	Requires strict operational controls regarding movement and hydraulic safety (Section 11).
Explosive Magazines	Temporary storage of seismic explosives at camp.	Must strictly separate detonators and main charges; must comply with CCE rules.

Equipment / System	Purpose	Standard Requirement / Context
Seismic Recording Gear	Captures reflected energy waves (geophones, cables, recording trucks).	Safe deployment procedures required during data acquisition (Section 9).

8. SAFETY REQUIREMENTS

- **Explosive Safety:** Meticulous inventory tracking and transport rules for seismic charges to prevent premature detonation.
- **Personnel Health:** Mandatory first aid training, designated doctors for the camp, pest control, and hygienic water storage.
- **Vehicle Safety:** Stringent rules (Section 10) for operating off-road vehicles carrying heavy loads over unpredictable terrain.

9. OPERATIONAL REQUIREMENTS

Field communication systems (Section 4) are a vital operational requirement. Given the remote nature of exploration, robust communication must be established between the base camp, drilling crews, shooting crews, and the recording station to ensure no detonations occur while personnel are in the danger zone.

10. INSPECTION, TESTING & MAINTENANCE

Section 15 of the standard mandates the use of extensive **Check Lists**. While specific checklist criteria require the full standard text, these lists are utilized by HSE personnel to systematically audit camp hygiene, explosive magazine compliance, vehicle condition, and drilling rig safety before and during operations.

11. EMERGENCY & HSE ASPECTS

The standard places a heavy emphasis on health in remote environments. Regular pest control (spraying) and fenced camps mitigate vector-borne diseases and wildlife attacks. Mandatory first-aid readiness and regular visits from a designated medical professional ensure that if an accident or illness occurs far from a hospital, immediate care is available.

12. IMPORTANT TECHNICAL CONCEPTS

Seismic Surveying: A method where an energy source (explosive or vibrator) creates artificial waves that travel into the earth. When these waves hit rock boundaries, they reflect back to the surface and are recorded. The time taken reveals the depth of the rock, helping map potential oil/gas traps.

Shot Hole: A pre-drilled vertical hole where explosive charges are placed below the weathering layer to maximize energy transfer into the solid earth.

Vibroseis: A non-explosive seismic energy source. Heavy trucks drop a baseplate to the ground and vibrate at specific frequencies (a "sweep").

13. IMPORTANT TERMS & ABBREVIATIONS

Term / Abbreviation	Meaning	Relevance
OISD	Oil Industry Safety Directorate	The regulatory body issuing the standard.
DGMS	Director General of Mines Safety	Statutory authority governing mine and oilfield safety.
CCE	Chief Controller of Explosives	Authority governing the storage and transport of explosives.
Dheki System	Manual drilling system	A traditional, labor-intensive method for shallow shot hole drilling.

14. RELATED STANDARDS & REGULATIONS

OISD-STD-181 explicitly notes that it functions alongside several critical statutory acts in India:

- **Mines Act** (regulated by DGMS)
- **Explosive Act** (regulated by CCE)
- **Indian Electricity Act**
- **Motor Vehicle Act**
- **Environment Protection Act & Indian Forest Act**

15. PRACTICAL USE CASES

Scenario 1: Designing a Field Camp in Assam

Relevant aspect: Camp Safety (Section 3).

Why it matters: To prevent disease and wildlife incidents, the camp manager references OISD-181 to erect proper fencing, ensure pest control (DDT spraying), and establish a hygienic potable water supply before deploying the 100-man seismic crew.

Scenario 2: Transitioning to Vibroseis in Desert Terrain

Relevant aspect: Vibroseis Operations (Section 11).

Why it matters: When a company chooses not to use explosives due to security risks, they deploy Vibroseis trucks. OISD-181 provides the mechanical and operational safety guidelines for running these heavy hydraulic machines in the field.

Scenario 3: Explosive Transportation

Relevant aspect: Explosives (Section 7).

Why it matters: When moving explosives from the magazine to the shot holes, the shooting crew follows OISD-181 to ensure detonators are never transported in the same compartment as the main seismic charges, averting accidental detonation.

16. KEY TAKEAWAYS

1. OISD-STD-181 is the primary safety standard for onshore geophysical (seismic) operations in India.
2. It covers the entirety of field operations: camp setup, drilling, explosives, shooting, and recording.
3. Camp hygiene, potable water, fencing, and first aid are strictly mandated.
4. The standard regulates the safe usage of both explosive shot holes and mechanical Vibroseis energy sources.
5. Explosives and detonators must be stored, handled, and transported securely and separately.
6. It relies heavily on established checklists for inspecting camps, vehicles, and rigs.
7. The standard complements, rather than replaces, CCE rules for explosives and DGMS rules for oilfields.
8. Vehicle safety is paramount due to the off-road, rugged nature of seismic data acquisition.

17. INFORMATION LIMITATIONS

- **Verified Information:** The exact section titles (1 through 16), the inclusion of both Vibroseis and explosive operations, camp safety protocols (fencing, pest control, water), and the regulatory relationship with DGMS/CCE.

- **Information Requiring Official Access:** Specific setback distances for explosive magazines, the detailed line items in the field checklists (Section 15), specific depth limits for shot hole drilling, and technical annexures (I & II).

Note: "Detailed requirement is not publicly available in the sources reviewed." No quantitative setback distances or checklist specifics have been fabricated.

18. SOURCES & REFERENCES

1. **Scribd Document Repository** - OISD-STD-181 Geophysical Operations Safety Standards (Document Extracts detailing Foreword, Index, Camp Safety, and Scope).
2. **Oil Industry Safety Directorate (OISD)** - Official Standards Index.
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Thek hai, I have added the disclaimer back to the title page and also included page numbers at the bottom right of each page for easy navigation. Here is the updated PDF file for **OISD-STD-181**: There is a file you can reference named "OISD-STD-181_Research_Summary_Numbered.pdf". Refer to this file by its name verbatim. [source: 1]OISD-STD-181 Geophysical Operations Safety Standards Industry: Oil & Natural Gas Category: Exploration (Upstream) - Geophysics Edition/Revision: 1st Edition / Currently Active Research Date: August 26, 2026 DISCLAIMER: This is an AI-generated research summary based on publicly available information. It is NOT an official OISD publication and must not be considered a replacement for the original OISD standard. The official standard should always be consulted for authoritative requirements. Page 1 of 8
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The standard covers safety aspects during the following activities:
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6. DETAILED TOPIC-WISE SUMMARY

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17. INFORMATION LIMITATIONS

- **Verified Information:** The exact section titles (1 through 16), the inclusion of both Vibroseis and explosive operations, camp safety protocols (fencing, pest control, water), and the regulatory relationship with DGMS/CCE.

- **Information Requiring Official Access:** Specific setback distances for explosive magazines, the detailed line items in the field checklists (Section 15), specific depth limits for shot hole drilling, and technical annexures (I & II).

Note: "Detailed requirement is not publicly available in the sources reviewed." No quantitative setback distances or checklist specifics have been fabricated.

18. SOURCES & REFERENCES

1. **Scribd Document Repository** - OISD-STD-181 Geophysical Operations Safety Standards (Document Extracts detailing Foreword, Index, Camp Safety, and Scope).
2. **Oil Industry Safety Directorate (OISD)** - Official Standards Index.
3. **Sun Petrochemicals** - International Competitive Bidding ICB Tender Document (Referencing OISD-181 compliance for survey crews).

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